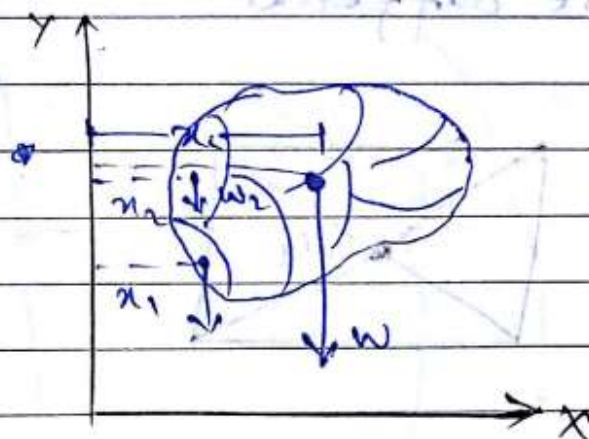


Centre Points

- i) Centre of Gravity
- ii) Centre of Mass
- iii) Centroid.



$$Wx_c = w_1x_1 + w_2x_2 + \dots + w_nx_n$$

$$x_c = \frac{w_1x_1 + w_2x_2 + \dots + w_nx_n}{W}$$

* If the body is small one, then the value of 'g' may be assumed uniform

Q COM ? COG ?

Centre of Mass

Centre of Gravity

• Independent of g

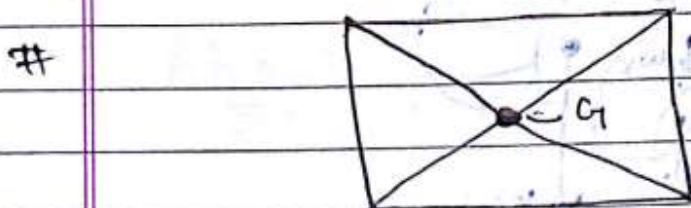
• Value depends on g

• Can Define in

• Case of free space

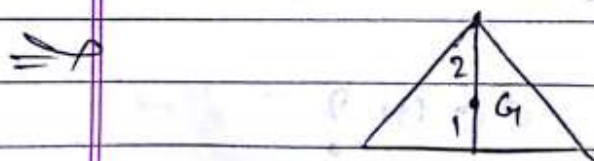
For a small body where 'g' is uniform COA & COM will be same.

For heavenly body COA & COM will be different.



Uniform thickness & uniform density, then all three COM , COA & Centroid will coincide.

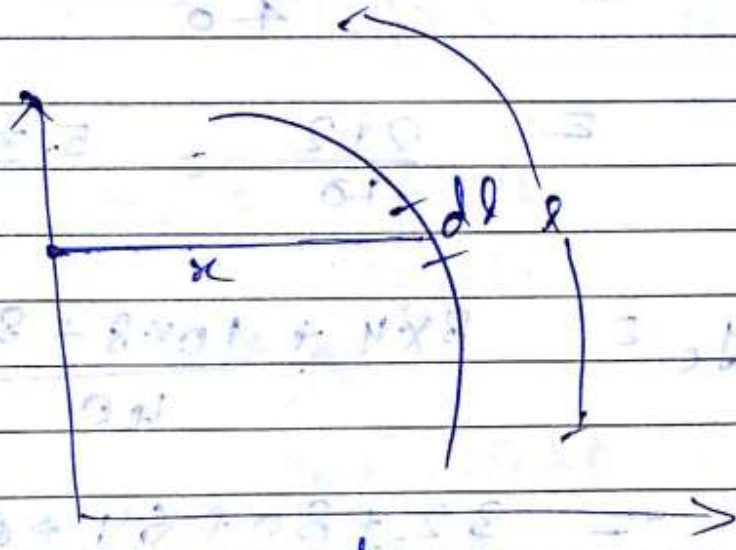
* For geometrical shapes, centroid Area, vol, length of rectangle, square, the centroid is the paramount



Centroid divides the median into 2:1.

$$y_c = \frac{w_1 y_1 + w_2 y_2 + \dots + w_n y_n}{w_1 + w_2 + \dots + w_n}$$

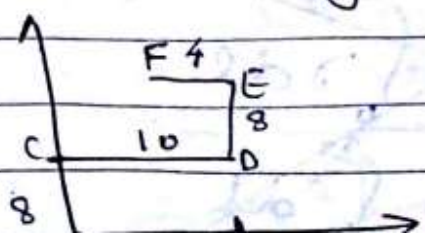
Q How to find C.G. of 1-D body?



$$x_c = \frac{\int_0^l x \, dl}{\int_0^l dl} \quad (\text{Abscissa})$$

$$y_c = \frac{\int_0^l y \, dl}{\int_0^l dl} \quad (\text{Ordinate})$$

Q1 Find the C.G. of bent up wire as shown in fig



Sol $x_c = \frac{10 \times 5 + (8 \times 0) + 10 \times 5 + 8 \times 10 + 4 \times 8}{10 + 8 + 10 + 8 + 4}$

$= \frac{50 + 50 + 80 + 32}{40}$

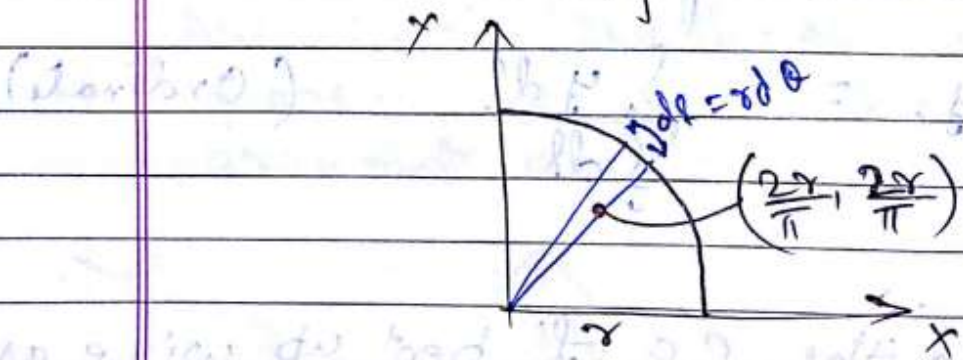
$= \frac{212}{40} = 5.3$

$y_c = \frac{8 \times 4 + 10 \times 8 + 8 \times 8 + 4 \times 16}{40}$

$= \frac{32 + 80 + 64 + 64}{40}$

$= 6.8$

Q2 Find the c.g. of a wire —



Sol $x_c = \frac{\int_0^{\pi/2} x \, dl}{\int_0^{\pi/2} dl}$

~~$$= \frac{\partial \int_0^{\pi/2} r \, d\theta}{\partial \pi r}$$~~

$$x_c = \frac{\int_0^{\pi/2} x \, d\theta}{\int_0^{\pi/2} d\theta}$$

$$= \frac{\int_0^{\pi/2} r^2 \cos \theta \, d\theta}{r \cdot \theta \Big|_0^{\pi/2}}$$

$$= \frac{r^2 [\sin \theta]_0^{\pi/2}}{r \cdot \theta \Big|_0^{\pi/2}}$$

$$= \frac{2r^2}{r \cdot \pi} = \frac{2r}{\pi}$$

$$y_c = \frac{\int_0^{\pi/2} y \, d\theta}{\int_0^{\pi/2} d\theta}$$

$$= \frac{\int_0^{\pi/2} r^2 \sin \theta \, d\theta}{r \cdot \theta \Big|_0^{\pi/2}}$$

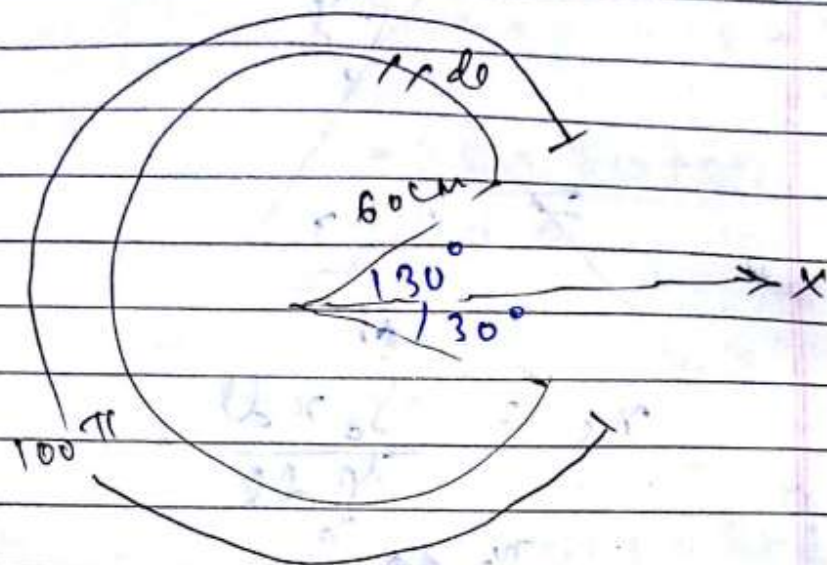
$$= \frac{2r^2 [-\cos \theta]_0^{\pi/2}}{r \cdot \pi} = \frac{2r}{\pi}$$

12/07/20

3/80

Date:	Neelgagan
Page No.	

Q3 Find the CG of the wire



$$x_c = \frac{\int_{\pi/6}^{5\pi/6} r d\theta \times r \cos \theta}{\int_{\pi/6}^{5\pi/6} r d\theta}$$

$$= \frac{r^2 [\sin \theta]_{\pi/6}^{5\pi/6}}{r \left(\frac{10\pi}{6} \right)}$$

$$= \frac{r \cdot \left[-\frac{1}{2} - \frac{1}{2} \right] 10\pi}{6}$$

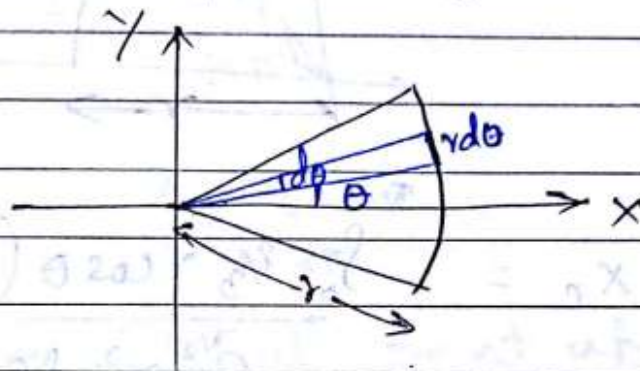
$$= -\frac{10\pi \times r}{6} = -\frac{5\pi r}{3}$$

for 2-D bodies

$$x_c = \frac{\int x dA}{\int dA}$$

$$y_c = \frac{\int y dA}{\int dA}$$

Q1 Find CG of the given sector —



$$dA = r d\theta \times r$$

$$x_c = \frac{\int_{-\alpha}^{\alpha} \frac{2}{3} r \cos\theta \frac{r d\theta \times r}{2}}{\int_{-\alpha}^{\alpha} \frac{r d\theta \times r}{2}}$$

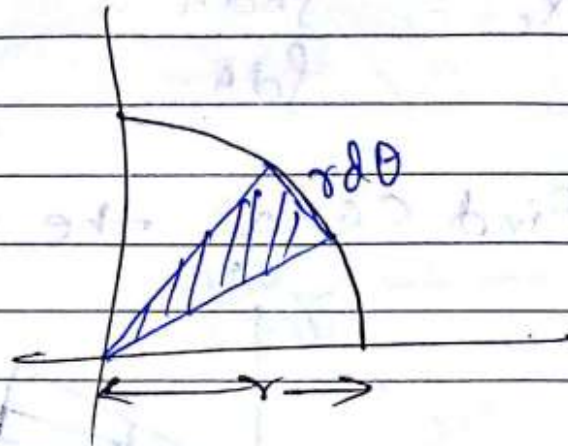
$$= \frac{r^3/3 \int_{-\alpha}^{\alpha} \cos\theta d\theta}{r^3/2 \int_{-\alpha}^{\alpha} d\theta}$$

$$= \frac{2r}{2} \frac{\sin\alpha}{\alpha}$$

$$y_c = 0$$

because symmetric about x-axis.

Q2 Find the CG of a quadrant of radius r



Sol
$$x_c = \frac{\int_0^{\pi/2} \frac{2}{3} r \cos \theta \left(\frac{r d\theta}{2} \times r \right)}{\int_0^{\pi/2} \frac{r^2}{2} d\theta}$$

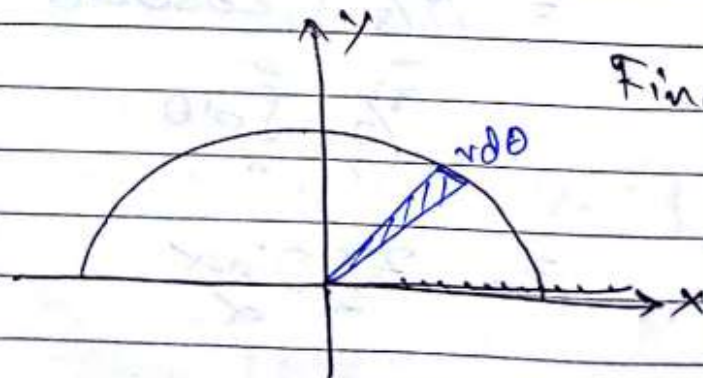
$$= \frac{4r}{3\pi}$$

Similarly

$$y_c = \frac{4r}{3\pi}$$

Q3

Find CG



As the fig is symmetric about y-axis $\therefore x_c = 0$

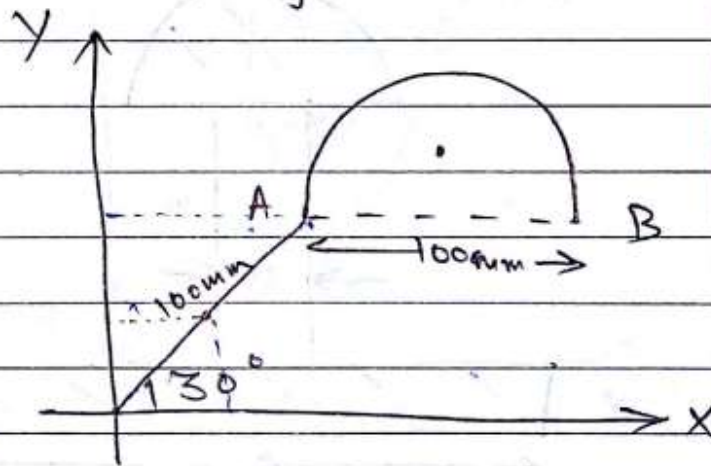
For γ_c

$$\gamma_c = \frac{\int_0^{2\pi} \frac{2}{3} r \sin \theta \left(\frac{r d\theta}{2} \times r \right)}{\pi \int_0^{2\pi} \frac{r^2}{2} d\theta}$$

$$= \frac{4r}{3\pi}$$

CG $\left(0, \frac{4r}{3\pi}\right)$

Q4 find a CG of bent up wire



$$X_c = \frac{l_1 x_1 + l_2 x_2}{l_1 + l_2}$$

$$= \frac{100 \times 50 \times \frac{\sqrt{3}}{2} + 100\pi \times (50\sqrt{3} + 50)}{100 + 50\pi}$$

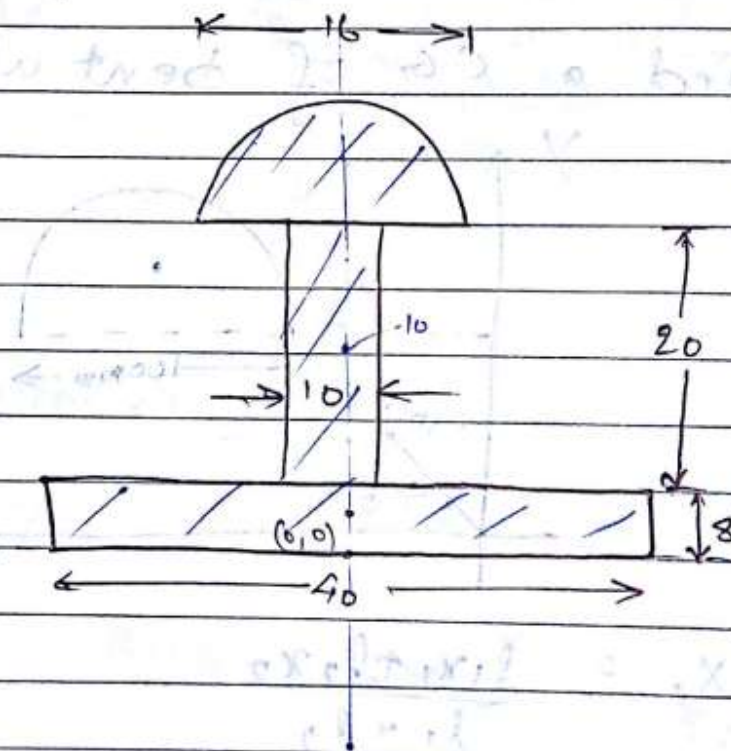
$$= \frac{4330 + 21430.5}{257} = 100.23$$

$$\bar{y}_c = \frac{y_1 l_1 + y_2 l_2}{l_1 + l_2}$$

$$= \frac{100 \times 25 + 50\pi \left(50 + \frac{200}{3\pi}\right)}{257}$$

$$= 59.72$$

Q5 CG of a composite shaded fig —



Sol:

As the fig is symmetric abt y-axis

$$\text{So } \bar{x}_c = 0$$

For

$$\bar{y}_c$$

$$\bar{y}_c = \frac{A_1 \bar{y}_1 + A_2 \bar{y}_2 + A_3 \bar{y}_3}{A_1 + A_2 + A_3}$$

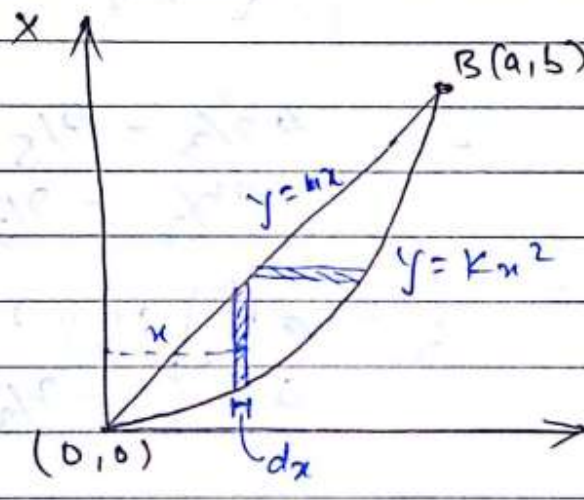
$$= \frac{320 \times 4 + 200 \times 18 + 32\pi \times \left[28 + \frac{32}{3\pi}\right]}{320 + 200 + 32\pi}$$

$$= \frac{1280 + 3600 + 3154.7}{620.48}$$

$$= 12.94$$

21.3.15

Q1



Sol

$$\bar{y}_c = \frac{\int_0^a x \left(\frac{b}{a}x - \frac{b}{a^2}x^2 \right) dx}{\int_0^a \left(\frac{b}{a}x - \frac{b}{a^2}x^2 \right) dx}$$

$$= \frac{\left[\frac{b}{a} \frac{x^3}{3} - \frac{b}{a^2} \frac{x^4}{4} \right]_0^a}{\left[\frac{b}{a} \frac{x^2}{2} - \frac{b}{a^2} \frac{x^3}{3} \right]_0^a}$$

$$= \frac{b a^2 \left(\frac{1}{3} - \frac{1}{4} \right)}{b a^2 \left(\frac{1}{2} - \frac{1}{3} \right)} = \frac{1}{4}$$

$$Y_c = \frac{\int_0^b y \left[\frac{a}{b} y - \sqrt{\frac{y}{b}} a \right] dy}{\int_0^b \left[\frac{a}{b} y - \sqrt{\frac{y}{b}} a \right] dy}$$

$$= \frac{\int_0^b \left(\frac{a}{b} y^2 - y^{3/2} \frac{a}{b^{1/2}} \right) dy}{\int_0^b \left(\frac{a}{b} y - y^{1/2} \frac{a}{b^{1/2}} \right) dy}$$

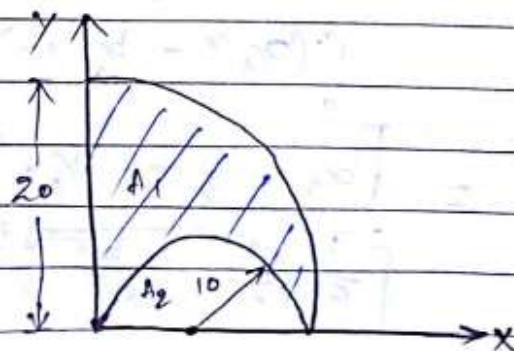
$$= \frac{\left[\frac{a}{b} \frac{y^3}{3} - \frac{2y^{5/2}}{5} \frac{a}{b^{1/2}} \right]_0^b}{\left[\frac{a}{b} \frac{y^2}{2} - \frac{2y^{3/2}}{3} \frac{a}{b^{1/2}} \right]_0^b}$$

$$= \frac{ab^2/3 - 2/5 b^2 a}{ab/2 - 2/3 ab}$$

$$= \frac{ab \left[1/3 - 2/5 \right]}{ab \left(1/2 - 2/3 \right)}$$

$$= b \times \frac{1/15}{1/6} = \frac{2b}{5}$$

Q2



Semicircle is removed from quadrant find the CG of remaining area.

Sol-
$$x_c = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2}$$

$$= \frac{\frac{1}{4} \pi (20)^2 \times \left(\frac{4 \times 20}{3\pi}\right) - \frac{1}{2} \pi (10)^2 \times 10}{\frac{1}{4} \pi (20)^2 - \frac{1}{2} \pi (10)^2}$$

$$= \frac{2666.7 - 1570}{157} = 6.98$$

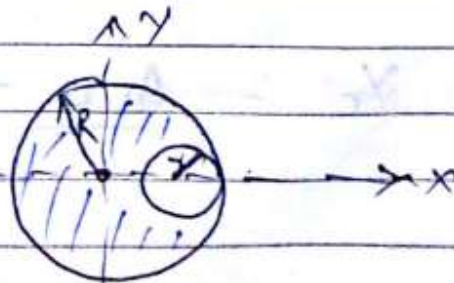
$$y_c = \frac{A_1 y_1 - A_2 y_2}{A_1 - A_2}$$

$$= \frac{\frac{1}{4} \pi (20)^2 \times \left(\frac{4 \times 10}{3\pi}\right) - \frac{1}{2} \pi (10)^2 \times \frac{4 \times 10}{3\pi}}{\frac{1}{4} \pi (20)^2 - \frac{1}{2} \pi (10)^2}$$

$$= \frac{2666.7 - 666.7}{157}$$

$$= 12.73$$

Q3 Find the CG of shaded region



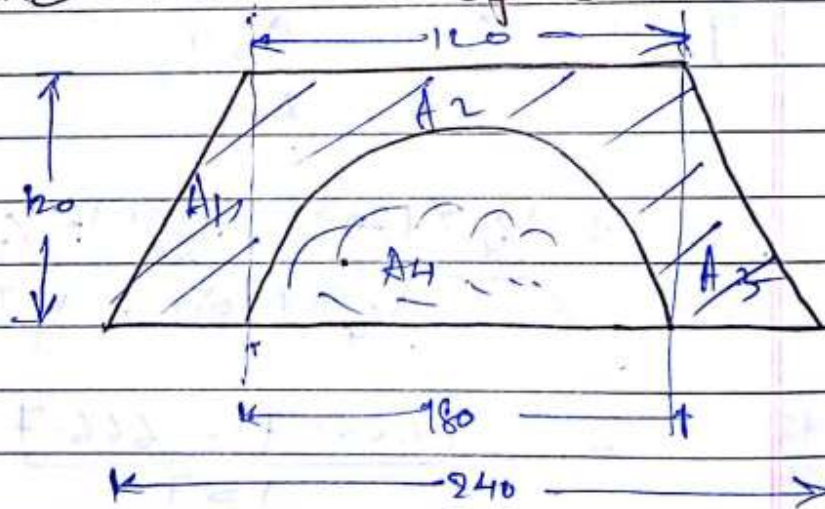
Sol Symmetric about X axis, So
 $y_c = 0$

$$x_c = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2}$$

$$= \frac{\pi R^2 \times R - \pi r^2 (2R - r)}{\pi R^2 - \pi r^2}$$

$$= \frac{R^3 - 2r^2 R + r^3}{R^2 - r^2}$$

Q4 Find the CA of shaded region



Sol - Symmetric about Y-axis
 So $x_c = 0$

$$y_c = \frac{A_1 y_1 + A_2 y_2 - A_3 y_3 - A_4 y_4}{A_1 + A_2 + A_3 - A_4}$$

$$= \cancel{\frac{1}{2} \times 120 \times 60 + 120 \times 120 + \frac{1}{2} \times 120 \times 60 - \frac{\pi \times}{2}}$$

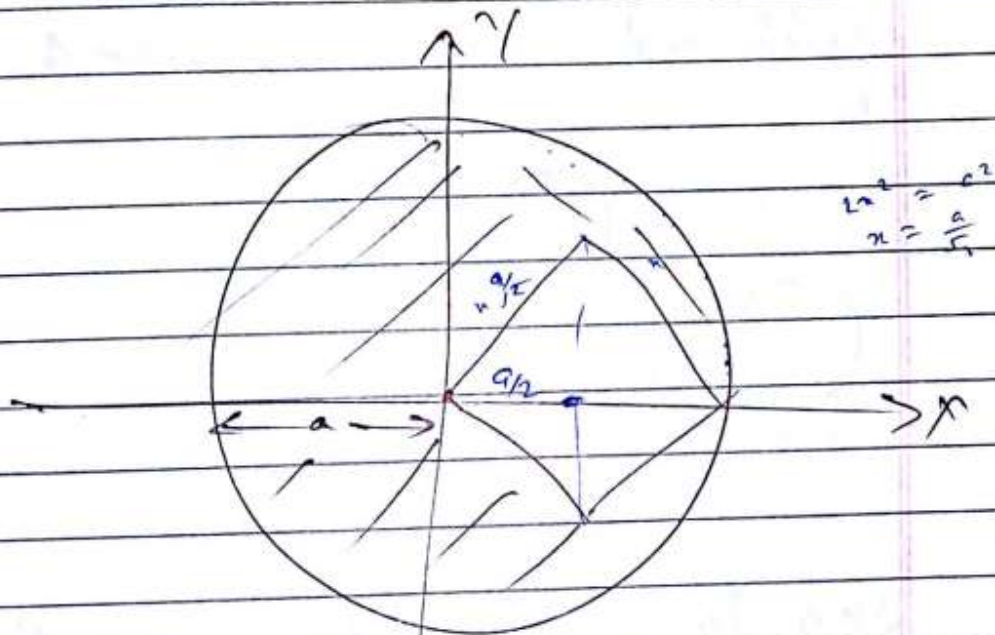
$$= \frac{1}{2} \times 120 \times 60 \times 2 \times 40 + 120 \times 120 \times 60 - \frac{\pi (90)^2 \left(\frac{4 \times 90}{3\pi} \right)}{2}$$

$$\frac{1}{2} \times 120 \times 60 \times 2 + 120 \times 120 - \pi (90)^2$$

$$= \frac{288000 + 864000 - 972000}{7200 + 14400 - 12717}$$

$$= \frac{180000}{8883} = 20.26$$

Q5



A square has been cut from a circle. Find CC of remaining area.

Sol As the fig is symmetric about x axis

$$y_c = 0$$

$$X_c = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2}$$

$$= \frac{\pi a^2 \times 0 - \pi \frac{a^2}{2} \times \frac{a}{2}}{\pi a^2 - \frac{a^2}{2}}$$

$$= \frac{-\frac{a^3}{4}}{\frac{a^2}{2} [2\pi - 1]}$$

$$= -\frac{a}{4 [2\pi - 1]}$$

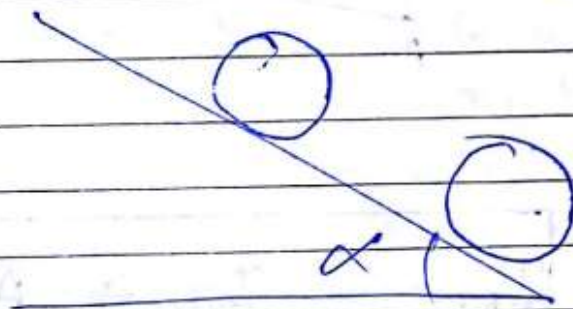
Moment of Inertia

Two types -

↳ Mass Moment of I

↳ Area Moment of I

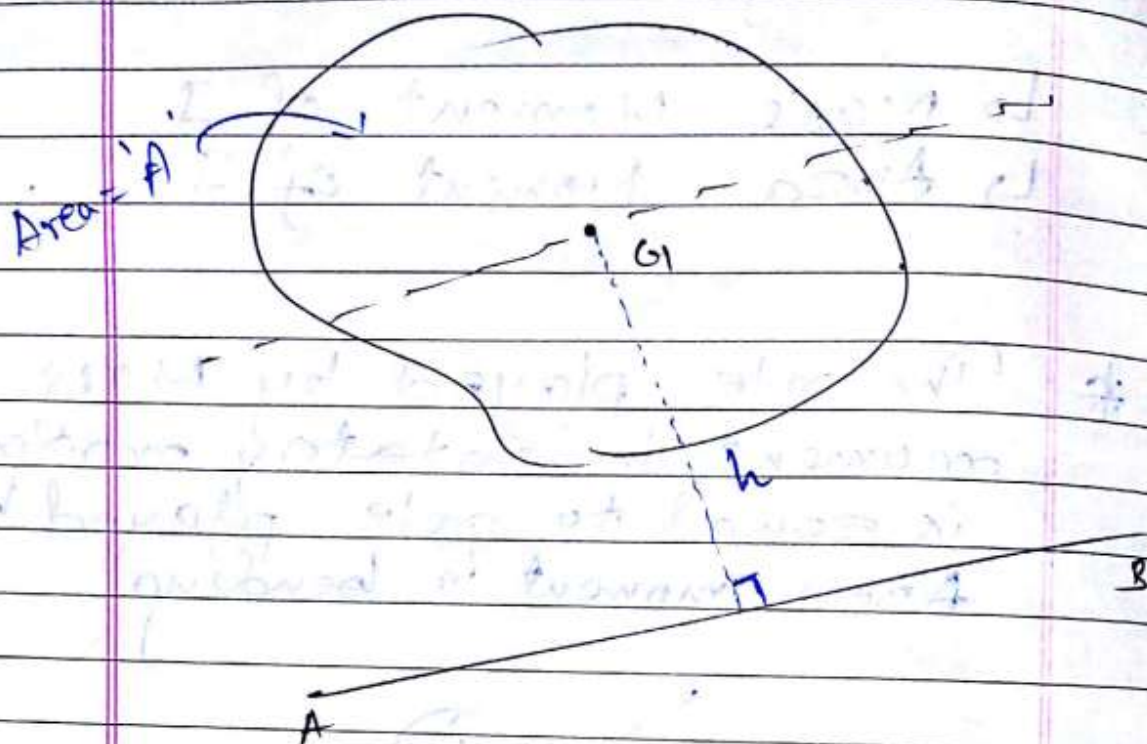
* The role played by Mass moment in rotatory motion is equal to role played by Area moment in bending



⇒ Theorem of Parallel axis

The moment of I of a lamina about any axis in the plane of the lamina equals to the sum of the moment of the inertia about the parallel central axis in the plane of the lamina and product of the

area of lamina & the sq of
the distance b/w two ~~laminar~~ axis



$$I_{AB} = I_G + Ah^2$$

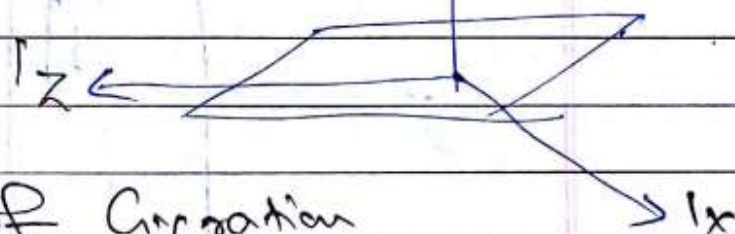
When $h=0$, then the
M.O.I will be min i.e
passing through centroid.

$\Rightarrow$ Theorem of perpendicular axis

The moment of I of an
area about an axis I to its
plane (polar moment of I) at a

pt. O is equal to the sum of moment of Inertia abt any 2 axis mutually $\perp$ through the same pt O & lying in the plane of area.

$$I_z = I_x + I_y$$



Radius of Gyration
M

The radius of Gyration K describes the way in which the area of ^{Cross-section} C.S. is distributed about its Centroidal Axis. If the Area of the lamina is assumed to be conc. at a distance of K ,

$$I = A K^2$$

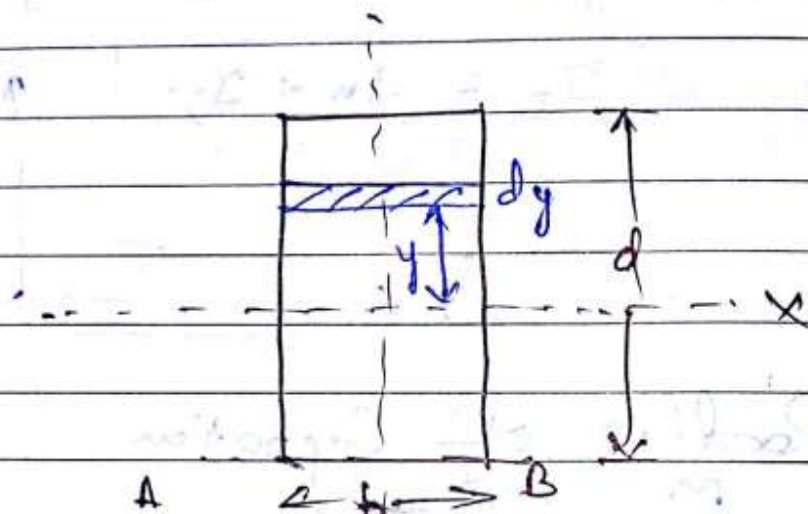
$$K = \sqrt{\frac{I}{A}}$$

$$\therefore K_x = \sqrt{\frac{I_x}{A}} \quad K_y = \sqrt{\frac{I_y}{A}}$$

(*) If value of K is more, it will have greater resistance

to buckling,

1. M.O.I of a Rectangle



$$\begin{aligned}
 I_x &= \int_{-d/2}^{d/2} y^2 (b dy) \\
 &= \left[\frac{y^3 b}{3} \right]_{-d/2}^{d/2} = \frac{d^3 b}{24} + \frac{d^3 b}{24} \\
 &= \frac{bd^3}{12}
 \end{aligned}$$

Q Area M.O.I is also known as 2nd Moment of the Area because in this case moment is taken of the m.o.i.

$$I_y = \int_{-b/2}^{b/2} x^2 (d dx)$$

$$= \left[\frac{x^3 d}{3} \right]_{-b/2}^{b/2}$$

$$d > b$$

So

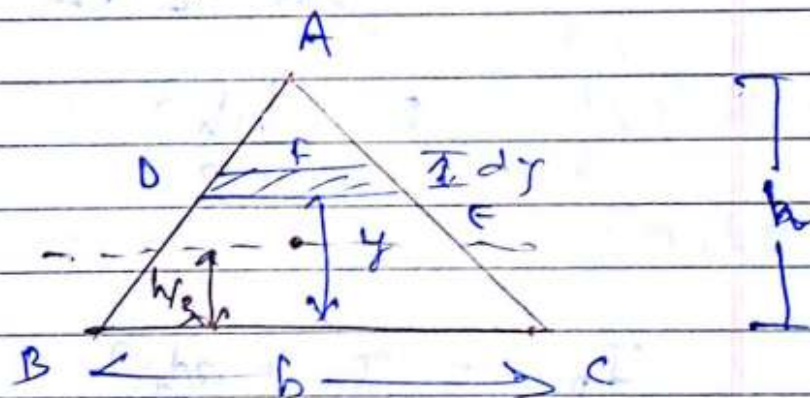
$$I_x > I_y$$

$$I_{AB} = I_x + A\left(\frac{d}{2}\right)^2$$

$$= \frac{bd^3}{12} + 3\frac{bd^3}{12}$$

$$= \frac{\cancel{3}bd^3}{12} = \frac{bd^3}{3}$$

2. M.O.I of a triangle



$$\frac{DE}{BC} = \frac{AF}{AZ}$$

$$\frac{DE}{b} = \frac{h-y}{h}$$

$$DE = \frac{b(h-y)}{h}$$

$$I_{Bc} = \int_0^h by^2 \cdot \frac{h-y}{h} dy$$

$$= \frac{b}{h} \left[\int_0^h y^2 h - y^3 dy \right]$$

$$= \frac{b}{h} \left[\frac{y^3 h}{3} - \frac{y^4}{4} \right]_0^h$$

$$= \frac{b}{h} \left[\frac{4h^3}{12} - \frac{3h^4}{12} \right]$$

$$= \frac{bh^3}{12}$$

$$I_{Bc} = I_a + a\left(\frac{h}{3}\right)^2$$

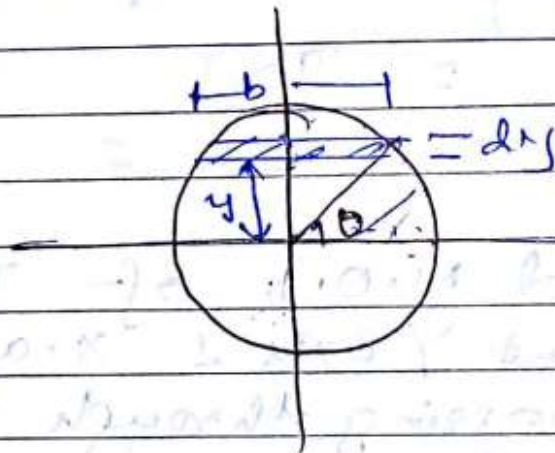
$$\frac{bh^3}{12} = I_a + \frac{1}{2} \times \frac{bh^3}{9}$$

$$I_a = \frac{bh^3}{12} - \frac{bh^3}{18}$$

$$\boxed{I_a = \frac{bh^3}{36}}$$

M.O.I of Various body

Circular Area .



$$I_x = \int y^2 dA$$

$$= \int y^2 b dy = \int_0^{2\pi} \int_0^R y^2 \sin^2 \theta dr d\theta$$

$$= 2 \int_{-\pi/2}^{\pi/2} r^2 \sin^2 \theta \times r \cos^2 \theta d\theta$$

$$= \frac{2r^4}{4} \int_{-\pi/2}^{\pi/2} (2 \sin \theta \cos \theta)^2 d\theta$$

$$= \frac{r^4}{2} \left[-\frac{\cos 2\theta}{2} \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{r^4}{2} \left[\frac{1}{2} + \frac{1}{2} \right]$$

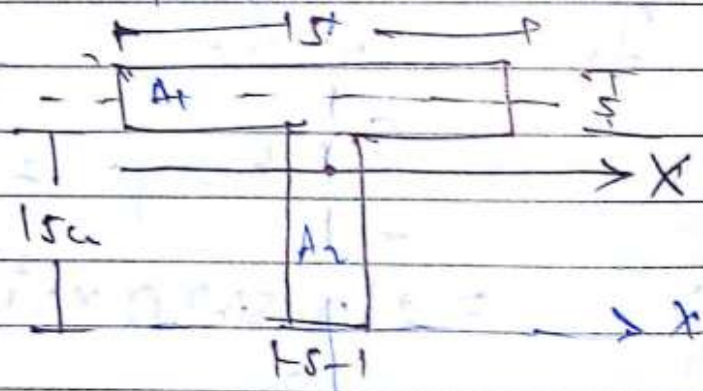
$$= \frac{\pi r^4}{4} = \frac{\pi d^4}{64}$$

$$I_y = \frac{\pi r^4}{4}$$

$$I_z = I_x + I_y$$

$$= \frac{\pi r^4}{2}$$

Q Find M.O.I of T section about Y axis & X-axis passing through the centroid of the section.



Sol. $x_c = 0$

$$y_c = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

$$= \frac{75 \times 7.5 + 7.5 \times 17.5}{150}$$

$$= 12.5 \text{ cm}$$

$$I_{xx} = \frac{15 \times 5^3}{12} + \frac{(15 \times 5) \times 5^2}{2}$$

$$= 15 \times 5^3 \left[\frac{1}{12} + 1 \right]$$

$$= \frac{15 \times 5^3 \times 13}{12 \times 4}$$

$$= 8125$$

$$I_{ay} = 8125 + \frac{5 \times 15^3}{12} + 15 \times 25$$

$$I_{ax} = \frac{15 \times 5^3}{12} + (15 \times 5) \times 5^2$$

$$+ \frac{5 \times 15^3}{12} + (15 \times 5) \times 5^2$$

$$I_{ay} = I_{ay1} + I_{ay2}$$

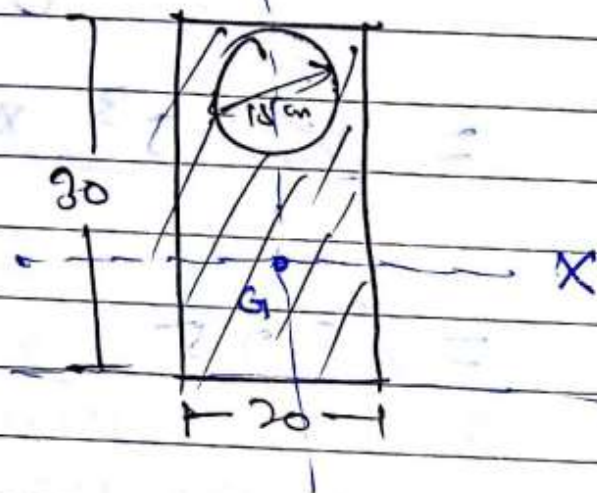
$$= \frac{5 \times 15^3}{12} + \frac{15 \times 5^3}{12}$$

$$= \frac{75}{12} [15^2 + 5^2]$$

$$= \frac{75}{12} \times [225 + 25]$$

$$= 1562.5 \text{ cm}^4$$

Q Find the M.O.I of a mono-rectangle plane as shown in fig about centroidal x & y axes.



24 Sol - 1. CG —

Symmetry about y -axis

$$x_c = 0$$

$$\begin{aligned}
 Y_c &= \frac{600 \times 15 - \pi \times \left(\frac{15}{2}\right)^2 \times \left(\frac{15}{2}\right)}{600 - \pi \left(\frac{225}{4}\right)} \\
 &= \frac{9000 - \pi \times \frac{225}{4} \times \left(\frac{45}{2}\right)}{600 - \frac{225\pi}{4}} \\
 &= \frac{5025.9375}{428.375}
 \end{aligned}$$

(Ar)

$$I_{xx} = \frac{20 \times 30^3}{12} + (20 \times 30) \times (3.14)^2 - \left[\frac{\pi (15)^4}{4} + \pi (15)^2 \times (10.6)^2 \right]$$

$$= 28387.4 \text{ cm}^4$$

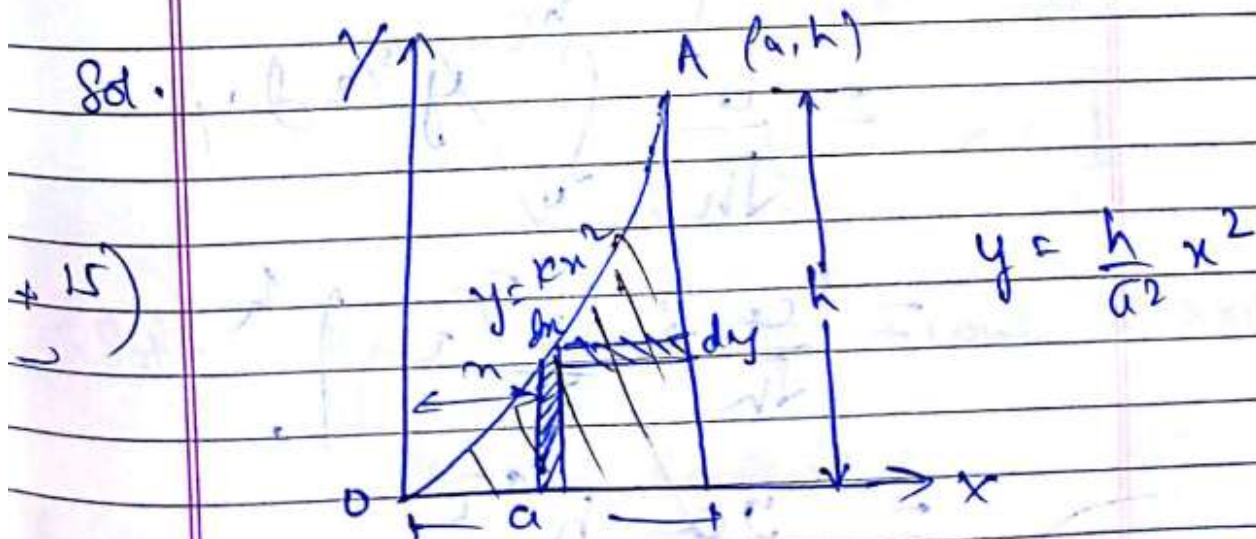
$$I_{yy} = \frac{30 \times 20^3}{12} - \frac{\pi (15)^4}{4}$$

$$= 20000 - 2483.7$$

$$= 17514.9$$

Q Find the M.O.I of shaded area

Sol.



$$\therefore h = K a^2$$

$$\Rightarrow K = \frac{h}{a^2}$$

$$I_{yy} = \int_0^a x^2 \left(\frac{hm^2}{a^2} x dx \right)$$

$$= \int_0^a \frac{h}{a^2} x^4 dx$$

$$= \frac{h}{a^2} \left[\frac{x^5}{5} \right]_0^a$$

$$= \frac{ha^3}{5}$$

$$I_{xx} = \int_0^h y^2 \left[a - a \sqrt{\frac{y}{h}} \right] dy$$

$$x^2 = \frac{a^2}{h}$$

$$x = a \sqrt{\frac{y}{h}}$$

$$= a \int_0^h y^2 - \frac{y^{5/2}}{\sqrt{h}} dy$$

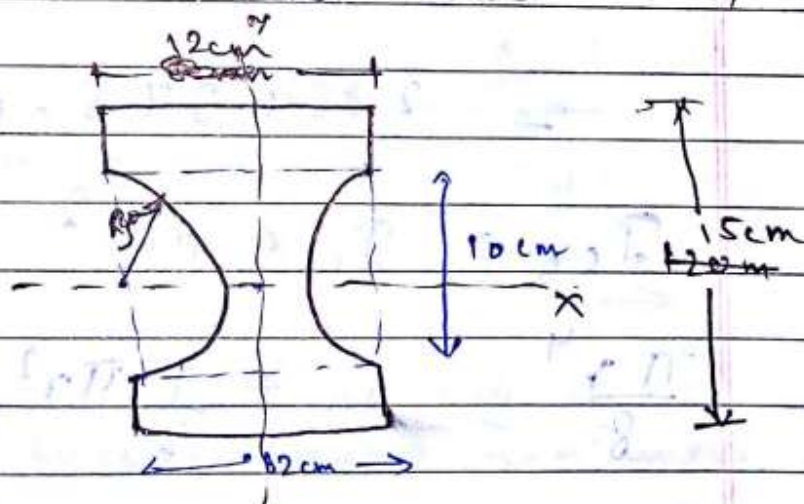
$$= a \left[\frac{y^3}{3} - \frac{2y^{7/2}}{7\sqrt{h}} \right]_0^h$$

$$= \frac{2ah^3}{21}$$

25/3/15

Imp

Q1 Find the M.O.I of the shaded area about centroidal x & y-axis



Sol: As Symmetry about x-axis & y-axis

$$X_c = 0 \quad Y_c = 0$$

For M.O.I —

$$I_{yy} = \cancel{2 \times (2.5)^3 \times 12} + \cancel{(7.5)^2 \times 30 \times 2}$$

$$= 15.625 +$$

$$= 31.25 + 3375 + 1000 = 490.625$$

$$= 3915.625 \text{ cm}^4$$

$$I_{xx} = \frac{12 \times (1.5)^3}{12} - 2 \times \left(\frac{\pi r^4}{8} \right)$$

$$= 3375 - 2 \times \pi \times 5 \times 5 \times 25$$

$$= 3375 - 490.625$$

$$= 2884.375 \text{ cm}^4$$

$$I_{CG} = I_C + Ah^2$$

$$\frac{\pi r^4}{8} = I_C + \frac{1}{2} \pi r^2 \left(\frac{42}{3\pi} \right)^2$$

$$I_C = \frac{\pi \times 5^4}{8} - \frac{1}{2} \times \frac{5^2 \times 4 \times 5^2 \times 4}{9 \times \pi}$$

$$= 245.3125 - 44.232$$

$$= 201.08 \text{ cm}^4 \quad 68.38 \text{ cm}^3$$

$$I_y = I_C + ah^2$$

$$= 201.08 + \pi \times 25 \times (6 - 0.016)$$

$$= \cancel{353.24} \cdot 212.54 \text{ cm}^4$$

M.O.I. of

$$I_y = \frac{15 \times 12^3}{12}$$

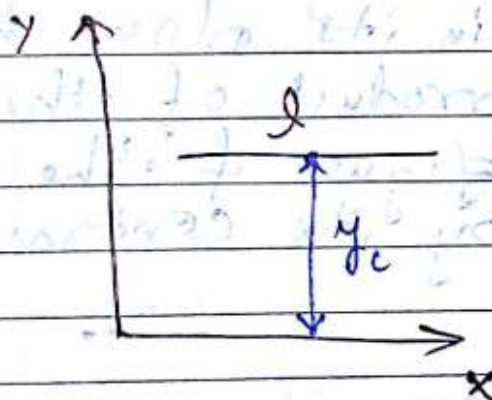
$$- 2 \left[\cancel{68.38} \cdot 212.54 \right]$$

$$= 1734.92$$

Pappus & ~~Cavalieri~~ Guldinus' Theorem

Theorem 1

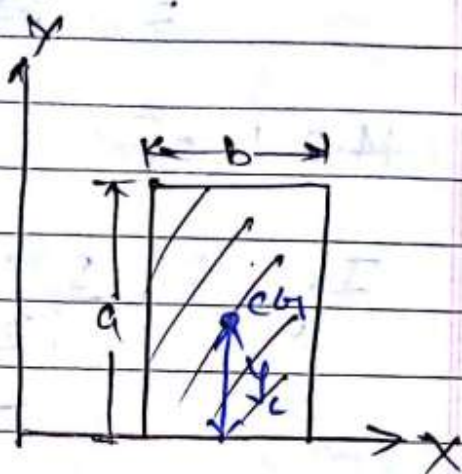
$$A = l \times 2\pi y_c$$



⇒ The surface area generated by ~~length l~~ by rotating a curve about a ~~known~~^{known} intersecting axis in its plane is equal to the product of the length of the curve and distance travelled by its centroid.

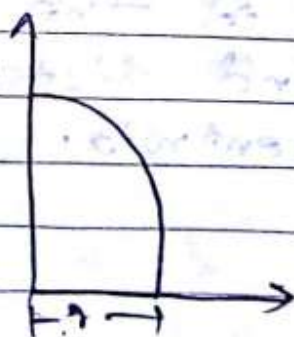
Theorem 2

$$V = A \times (2\pi y_c)$$



The volume of a solid generated by rotating any plane fig. about a non-intersecting axis in its plane is equal to the product of the area of the figure & the distance travelled by its centroid.

Q2 By using P & T Calculate the area of the surface of revolution by revolving a quarter of a circular arc of radius r about y axis.

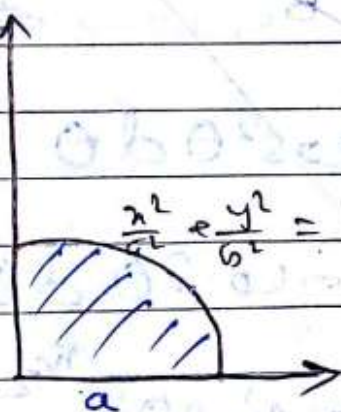


$$A = \frac{\pi \times r}{2} \times \cancel{2} \times 2\pi \times \frac{2r}{\pi}$$

$$= \cancel{\frac{2r^2}{3}} = \frac{4\pi r^2}{3}$$

$$= 2\pi r^2$$

Q3 By using PGT, calculate the vol. generated by revolving a quarter of the ellipse about x-axis



$$V = \frac{\pi ab}{4} \times 2\pi \times y_c$$

$$\therefore y_c = \frac{\int_0^b y \left(\frac{a}{b} \sqrt{b^2 - y^2} dy \right)}{\int_0^b \frac{a}{b} \sqrt{b^2 - y^2} dy}$$

$$= -\frac{a}{b} \int_0^b y \sqrt{b^2 - y^2} dy$$

$$b^2 - y^2 = t$$

$$-2y dy = dt$$

$$= 4b$$

$$\begin{aligned} V &= \frac{\pi a b \times 2\cancel{a} \times \cancel{4b}}{\cancel{3}\cancel{a}} \\ &= \frac{2\pi a b^2}{3} \end{aligned}$$